1) A carrier of 2 MHz has 1 kW of its power amplitude modulated with a sinusoidal signal of 2 KHz. The depth of modulation is 60%. Calculate the sideband frequencies, the signal bandwidth, the power in the sidebands and the total power in the modulated wave.

$$P_c = 1\,kW, \quad f_c = 2\,MHz, \quad f_m = 2\,KHz$$

$m = 0.6$

$$f_{LSB} = f_c - f_m = 2\,MHz - 2\,KHz$$
$$= 2000\,KHz - 2\,KHz$$
$$= 1998\,KHz = 1.998\,MHz$$

$$f_{USB} = f_c + f_m = 2\,MHz + 2\,KHz$$
$$= 2000\,KHz + 2\,KHz$$
$$= 2002\,KHz$$

$$BW = 2f_m = 2 \times 2\,KHz = 4\,KHz$$

Power in each side band is

$$P_{LSB} = P_{USB} = \frac{m^2}{4} P_c = \frac{0.6^2}{4} \times 1\,kW$$
$$= 0.09\,kW \text{ or } 90\,W$$

Total power in the AM is

$$P_t = P_c\left(1 + \frac{m^2}{2}\right) = 1\,kW\left(1 + \frac{0.6^2}{2}\right)$$
$$= \underline{1.18\,kW}$$

observe that when the depth of modulation is 60%. out of the total power of 1180W, only 90W is related in useful information as confined with sidebands.

2). Maximum and Minimum amplitudes of an AM wave are 600mV and 200mV respectively. Find the modulation index.

$V_{max} = 600mV$, $V_{min} = 200mV$

$$m = \frac{V_{max} - V_{min}}{V_{max} + V_{min}} = \frac{600-200}{600+200}$$

$$= 0.5 \text{ or } 50\%.$$

3) A carrier wave with amplitude 10V and frequency 10MHz is amplitude modulated by an audio signal of frequency 1kHz. Write the equation for this AM wave and sketch its frequency spectrum. Assume $m = 0.5$.

Given

$V_c = 10V$, $f_c = 10MHz = 10 \times 10^6 Hz$

$f_m = 1kHz$, $m_a = 0.5$

$$v = V_c(1 + m \sin w_m t) \sin w_c t$$

$$w_c = 2\pi f_c = 2\pi \times 10 \times 10^6 = 62.8 \times 10^6 \text{ rad/s}$$

$$w_m = 2\pi f_m = 2\pi \times 1 \times 10^3 = 6280 \text{ rad/s.}$$

$$\underline{\underline{v = 10(1 + 0.5 \sin 6280t) \sin(62.8 \times 10^6 t)}}$$

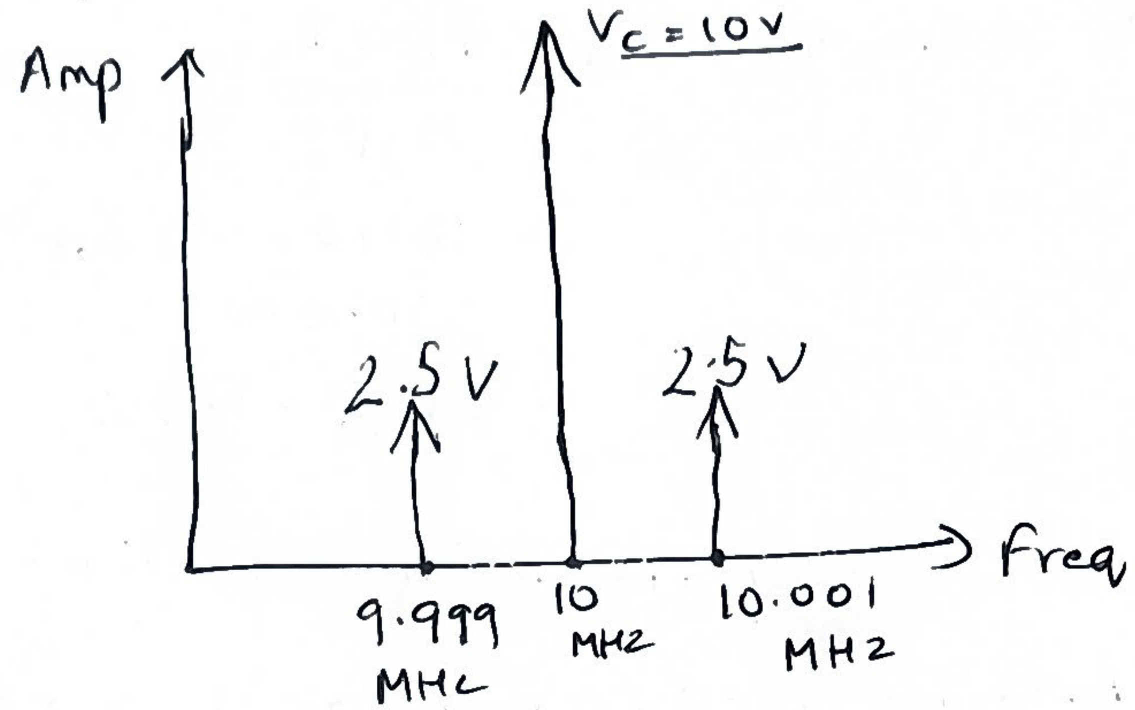
The amplitude of each sideband is $= \frac{mV_c}{2}$

$$= \frac{0.5 \times 10}{2}$$

$$= \underline{\underline{2.5V}}$$

$f_{LSB} = f_c - f_m = 10 \times 10^6 - 1 \times 10^3 = 9.999 \times 10^6$ Hz

$f_{USB} = f_c + f_m = 10 \times 10^6 + 1 \times 10^3 = 10.001 \times 10^6$ Hz



4. An AM Broadcasting station broadcasts with an average transmitted power of 20kW. Modulation index is set at 0.8. Find the saving in power if the power contained in the LSB alone is used.

Given $P_t = 20kW$, $m = 0.8$.

Carrier Power $P_c = \dfrac{P_t}{1 + \frac{m^2}{2}} = \dfrac{20}{1 + \frac{0.8^2}{2}}$

$= 15.15$ kW.

The power in the sidebands is

$P_t - P_c = 20 - 15.15 = 4.85$ kW.

The power in the LSB is

$P_{LSB} = \dfrac{4.85}{2} = 2.425W$.

Saving in power is $P_t - P_{LSB} = \overset{20}{\cancel{15.15}} - 2.425$

$= \cancel{17.575\,kW} = 17.575\,k$

$$\% \text{ saving in Power} = \frac{P_t - P_{LSB}}{P_t} \times 100\%$$

$$= \frac{17.575}{20} \times 100\%$$

$= \cancel{88\%}\ \underline{\underline{87.9\%}}$

Superheterodyne Receiver

1) Antenna receives many radio signals.

2) RF stage selects one station.

3). Local oscillator generates another frequency